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Math 215A

Homework 5
Due: October 28, 2026

Rings are understood to be commutative, unless stated otherwise.

Problem 1

Give an example of a ring A and a multiplicatively closed subset $S \subset A$ such that the relation "$(a, s) \sim (b, t)$ if $at - bs = 0 \in A$" on $A \times S$ is not an equivalence relation. [This explains the more complicated definition of $A[S^{-1}]$, in general. Make sure you know that definition.]

Solution. Consider $A = \mathbb{Z}_6$ and $S = \{1, 3\}$. Consider $(0, 3), (0, 1), (2, 3)$.

$$(0) * 1 - 3 * 0 = 0 \Rightarrow (0, 3) \sim (0, 1)$$
$$0 * 3 - 3 * 2 = 6 = 0 \Rightarrow (0, 3) \sim (2, 3)$$
$$0 * 3 - 1 * 2 = -2 = 4 \neq 0 \Rightarrow (0, 1) \text{ is not equivalent to } (2, 3).$$

Thus $\sim$ is not transitive. Hence it cannot be a equivalence relation.

Problem 2

Let A be a ring, $f \in A$, and M an A-module. Show that the localization $M[1/f]$ is isomorphic to the direct limit

$$L = \lim_{\rightarrow} \left(M \xrightarrow{f} M \xrightarrow{f} M \xrightarrow{f} \cdots \right).$$

Proof: Let $\{M_i\}_{i=1}^{\infty}$ be a collection of copies of M and for $x_i \in M_i$, x_i is a copy of $x \in M$ that is in M_i.

$$L = \bigsqcup_{i=1}^{\infty} M_i / \left(x_i \sim y_j \Leftrightarrow \exists k \geq i, j \ni f^{k-i} x_i = f^{k-j} y_j \right)$$

Let

$$T : L \to M\left[\frac{1}{f}\right]$$
$$x_i \to \frac{x_i}{f^i}$$

Well defined: Suppose $x_i \sim y_j$, thus exists $k \geq i, j$ such that

$$f^{k-i} x_i = f^{k-j} y_j.$$

Furthermore, note that $x_i \sim f^{k-i} x_i = f^{k-j} y_j \sim y_j$ and that $f^{k-i} x_i = f^{k-j} y_j \in M_k$.

$$T\left(f^{k-i} x_i\right) = \frac{f^{k-i} x_i}{f^k} = \frac{x_i}{f^i} = T(x_i)$$

Thus $T\big(f^{k-i}x_i\big) = T(x_i)$. By a symmetric argument, $T\big(f^{k-j}y_j\big) = T\big(y_j\big)$. Thus

$$T(x_i) = T\big(f^{k-i}x_i\big) = T\big(f^{k-j}y_j\big) = T\big(y_j\big).$$

Thus T is well defined.

A linear: Let $k \geq i, j$ thus

$$T\big(x_i + y_j\big) = \frac{f^{k-i}x_i + f^{k-j}y_j}{f^k} = \frac{x}{f^i} + \frac{y}{f^j} = T(x_i) + T\big(y_j\big)$$

and

$$T(ax_i) = \frac{ax}{f^i} = aT(x_i)$$

Injective: Suppose $T(x_i) = 0$. Then

$$0 = \frac{x}{f^i} \Rightarrow \frac{0}{f} = \frac{x}{f^i}.$$

Thus there exists a $n \in \mathbb{N}$ such that $f^n\big(fx - f^i * 0\big) = 0 \Rightarrow f^{n+1}x = 0 \Rightarrow x_i = 0 \in L$.

Surjective: $\forall r \in M\left[\frac{1}{f}\right]$,

$$r = \frac{m}{f^k}$$

for $m \in M$ and $k \in \mathbb{N}_0$. $T(m_k) = \frac{m}{f^k}$.

Thus the map is a isomorphism as a $A\left[\frac{1}{f}\right]$ module. □

Problem 3

Let R be a ring, $f \in R$. Show that the localization $R[1/f]$ is isomorphic to $R[x]/(fx-1)$ (as an R-algebra).

Proof: Let $T : R[x] \to R\left[\frac{1}{f}\right]$ where for $r \in R$, $r \to r$, and $x \to \frac{1}{f}$. T is a R-algebra homomorphism. Note that

$$\begin{aligned} 0 = T\left(\sum_{k=0}^{n} a_k x^k\right) &= \sum_{k=0}^{n} \frac{a_k}{f^k} \\ &= \frac{\sum_{k=0}^{n} f^{n-k}a_k}{f^k} \\ f^m\left(\sum_{k=0}^{n} a_k\right) &= \sum_{k=0}^{n} f^{n+m-k}a_k \\ &= 0 \end{aligned}$$

□

Problem 4

Let R be a ring, $S \subset R$ a multiplicatively closed subset, M an R-module. Suppose that for every $s \in S$, multiplication by s is a bijection from M to itself. Show that the natural map $M \to M[S^{-1}]$ is an isomorphism. As a sample application, show that the localization of the $\mathbb{Z}$-module $\mathbb{Z}/8$ at the prime ideal $(2) \subset \mathbb{Z}$ is simply $\mathbb{Z}/8$.

Problem 5

A *finite algebra* over a ring A means an A-algebra which is finitely generated as an A-module. Show that a finite flat $\mathbb{Z}$-algebra is free as a $\mathbb{Z}$-module, and give an example of a flat $\mathbb{Z}$-algebra which is not free as a $\mathbb{Z}$-module. Is a flat $\mathbb{Z}$-algebra of finite type always free as a $\mathbb{Z}$-module?